

**AI/ML Team – Member 4 Report****3D Avatar Generation and Body Reconstruction for Virtual Try-On Systems****Introduction**

Virtual Try-On technology is transforming the fashion industry by allowing users to visualize clothing on digital representations of themselves before making a purchase. Traditional 2D virtual try-on systems can provide a basic preview of garments, but they often fail to accurately represent body shape, pose, and garment fitting. To overcome these limitations, researchers have developed advanced 3D Avatar Generation and Body Reconstruction techniques.

3D Avatar Generation focuses on creating a digital representation of a human body, while Body Reconstruction involves recovering the three-dimensional structure of a person from images or videos. These technologies enable more realistic garment fitting, improved user experience, and higher accuracy in virtual fashion applications.

This report explores major body reconstruction models including SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting, along with their applications in modern virtual try-on systems.

**1. SMPL (Skinned Multi-Person Linear Model)**

SMPL is one of the most widely used parametric human body models in computer vision and graphics. It represents the human body as a 3D mesh consisting of thousands of vertices and allows modifications through body shape and pose parameters.

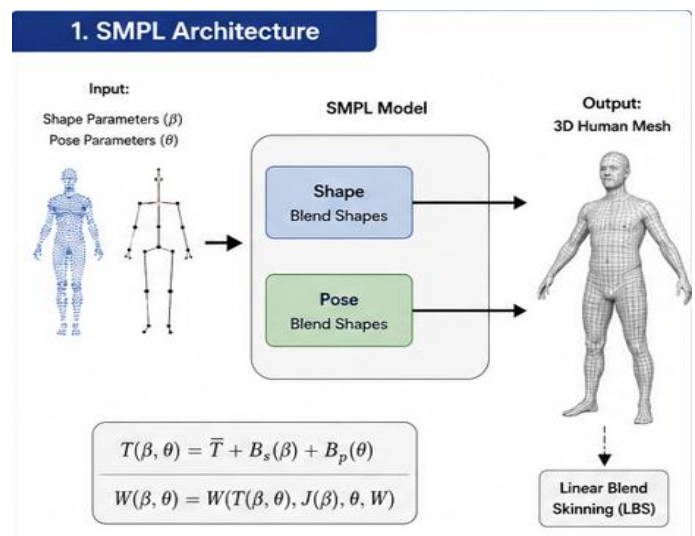
The model generates realistic human body structures by adjusting predefined parameters. Researchers and developers use SMPL as a foundation for various avatar generation and virtual fitting applications.

**Advantages**

- Lightweight and computationally efficient.
- Easy integration with other AI systems.
- Suitable for real-time applications.
- Provides consistent body structure representation.

**Limitations**

- Limited clothing details.
- Cannot accurately reconstruct loose garments.
- Produces simplified body shapes compared to modern methods.



## Applications

- Virtual fitting rooms.
- Motion capture systems.
- Human pose estimation.
- Gaming avatars.

## 2. PIFuHD (Pixel-Aligned Implicit Function for High Resolution Reconstruction)

PIFuHD is a deep learning model designed to reconstruct detailed 3D human models from a single RGB image. Unlike traditional mesh-based methods, PIFuHD captures fine geometric details such as hair, clothing folds, and body contours.

The model uses pixel-aligned implicit functions to estimate the occupancy of 3D points and generate highly detailed human reconstructions.

### Advantages

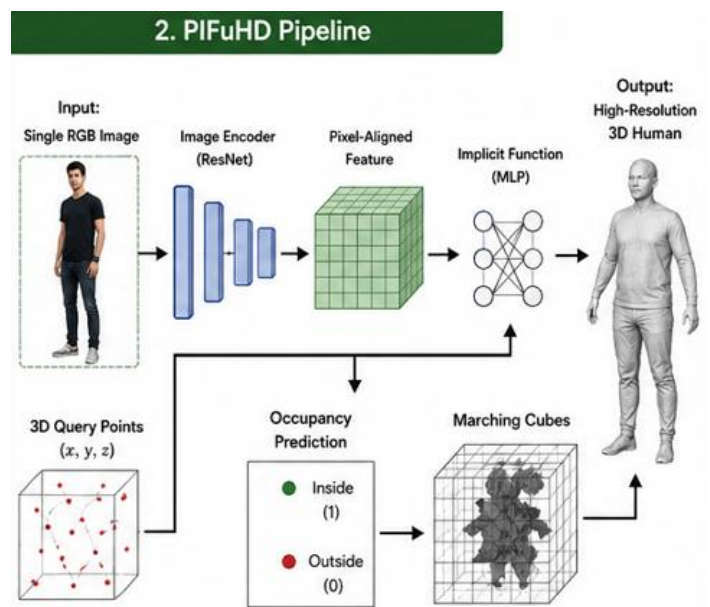
- Produces high-resolution 3D models.
- Preserves detailed body features.
- Works with a single image input.
- Generates realistic human representations.

### Limitations

- Computationally expensive.
- Longer processing time.
- Requires significant GPU resources.

### Applications

- Digital avatar creation.
- Fashion visualization.
- Virtual reality environments.
- Character generation.



## 3. ICON (Implicit Clothed Humans Obtained from Normals)

ICON is a modern body reconstruction framework specifically designed to reconstruct clothed human bodies. Unlike previous approaches that focus mainly on body shape, ICON preserves garment information and clothing geometry.

The model combines normal maps with implicit representations to generate realistic clothed avatars suitable for virtual try-on applications.

### Advantages

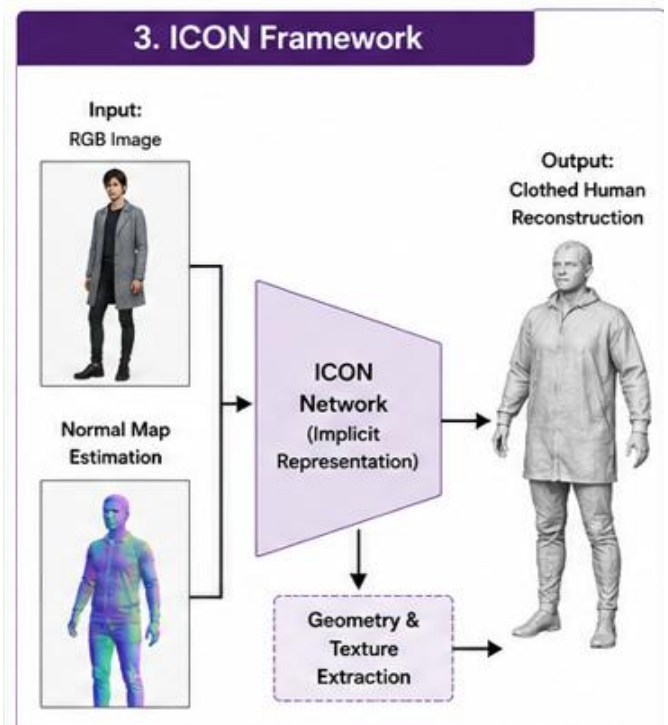
- Preserves clothing details effectively.
- Produces realistic clothed human models.
- Suitable for fashion applications.
- High reconstruction quality.

### Limitations

- Requires powerful hardware.
- More complex architecture.
- Increased computational cost.

### Applications

- Fashion technology.
- Virtual try-on systems.
- Clothing visualization.
- Digital fashion platforms.



## 4. HumanNeRF

HumanNeRF extends Neural Radiance Fields (NeRF) technology to dynamic human reconstruction. It creates highly realistic 3D representations of humans by learning appearance and geometry from multiple viewpoints.

The model captures body movements, clothing deformations, and fine visual details, resulting in photorealistic avatars.

### Advantages

- Extremely realistic rendering.
- Captures dynamic body movements.
- Preserves appearance details.
- High visual quality.

### Limitations

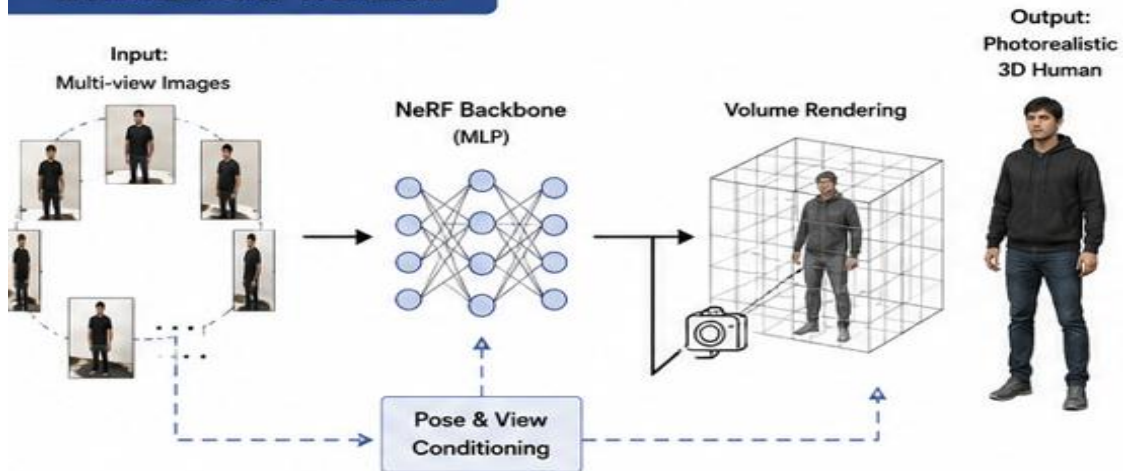
- Requires multiple images or videos.
- Training process is computationally intensive.
- Longer reconstruction time.

### Applications

- Virtual humans.
- Metaverse environments.

- Motion synthesis.
- Advanced virtual try-on systems.

#### 4. HumanNeRF Workflow



#### 5. 3D Gaussian Splatting

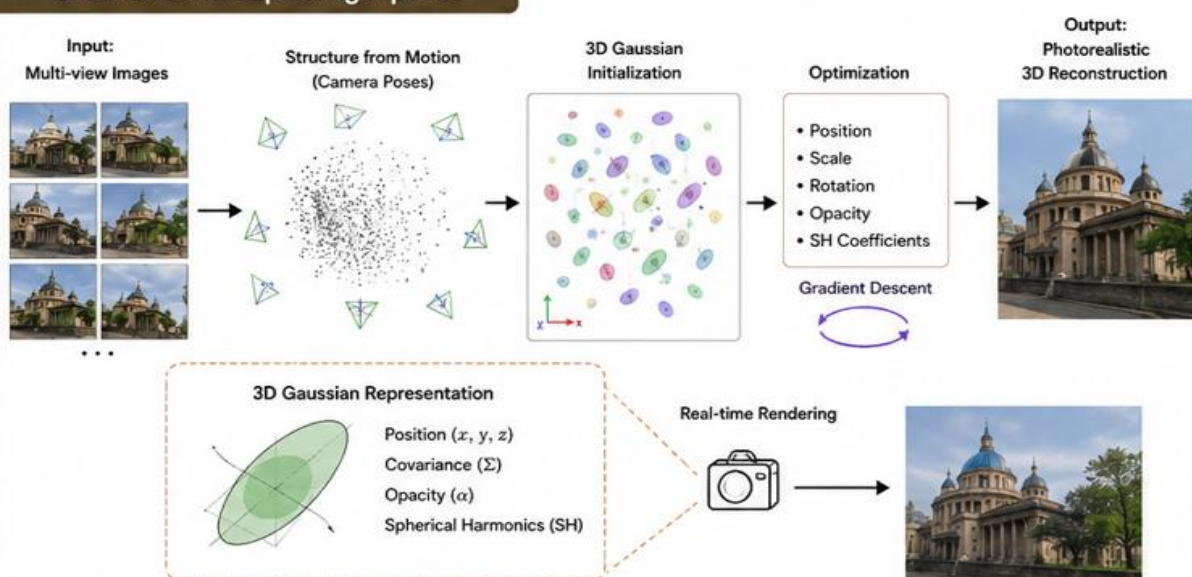
3D Gaussian Splatting is a recent technique for real-time photorealistic scene reconstruction and rendering. Instead of using traditional meshes, it represents objects using millions of Gaussian primitives.

This approach enables high-quality rendering while maintaining interactive performance.

##### Advantages

- Excellent visual quality.
- Real-time rendering capabilities.
- Accurate scene representation.
- Fast visualization after reconstruction.

#### 5. 3D Gaussian Splatting Pipeline



**Limitations**

- Large storage requirements.
- Complex implementation.
- Requires multiple views for best performance.

**Applications**

- Virtual reality.
- Augmented reality.
- Digital fashion visualization.
- Realistic avatar rendering.

**Comparison of Models**

| Model                 | Input                    | Output                           | Advantages                 | Limitations      |
|-----------------------|--------------------------|----------------------------------|----------------------------|------------------|
| SMPL                  | Images/Measurements      | 3D Body Mesh                     | Fast, lightweight          | Less detailed    |
| PIFuHD                | Single RGB Image         | High Resolution 3D Human         | Detailed reconstruction    | High computation |
| ICON                  | RGB Image                | Clothed Human Model              | Preserves clothing details | GPU intensive    |
| HumanNeRF             | Multi-view Images/Videos | Dynamic Human Model              | Realistic rendering        | Slow training    |
| 3D Gaussian Splatting | Multi-view Images        | Photorealistic 3D Representation | Real-time rendering        | Large storage    |

**Applications in Virtual Try-On Systems**

Modern virtual try-on platforms use 3D avatars to improve garment fitting accuracy. The workflow generally begins with capturing a user image, followed by body reconstruction and avatar generation. The selected garment is then digitally fitted onto the generated avatar.

Benefits include:

- Improved online shopping experience.
- Reduced product returns.
- Better size recommendations.
- Realistic clothing visualization.
- Personalized fashion suggestions.

These technologies are increasingly being integrated into e-commerce platforms, virtual fitting rooms, augmented reality shopping applications, and metaverse fashion environments.

### **Findings and Observations**

During the research, it was observed that different models serve different purposes. SMPL remains an excellent choice for lightweight applications and rapid avatar generation. PIFuHD provides superior geometric detail but requires more computational resources. ICON is particularly valuable for fashion applications because it preserves clothing information effectively. HumanNeRF generates highly realistic dynamic avatars suitable for immersive environments. 3D Gaussian Splatting represents one of the most promising recent developments due to its ability to produce photorealistic results with real-time rendering performance.

For virtual try-on systems, ICON and HumanNeRF provide the most relevant capabilities because they maintain garment details while generating realistic human representations. However, SMPL continues to be widely adopted due to its simplicity and efficiency.

### **Conclusion**

3D Avatar Generation and Body Reconstruction are essential components of modern virtual try-on systems. They bridge the gap between traditional online shopping and realistic garment visualization. Technologies such as SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting have significantly improved the quality of digital avatars and clothing simulations.

Among the studied methods, ICON appears particularly suitable for fashion-focused virtual try-on applications due to its ability to reconstruct clothed human bodies accurately. Future developments in this field are expected to focus on faster processing, improved realism, and real-time garment fitting capabilities, enabling more immersive and accurate virtual shopping experiences.